Compress the Relay, Steer the Reader

H-OBF relay compression × Judger-only concision steering (ASC) in LatentMAS

Sagnik Biswas

July 2026

Qwen3-14B · GSM8K · verified LatentMAS fork

The setup and the claim

[Question] $\rightarrow$ Planner $\rightarrow$ Critic $\rightarrow$ Refiner $\rightarrow$ **Judger** $\rightarrow$ [Answer]

- Planner/Critic/Refiner run 40 latent steps, emit **no text**; they grow a shared **KV cache** ($\sim$640 positions, $\sim$100 MB) — the entire inter-agent "message."
- Only the **Judger** decodes text.

Systems claim under test

Can a **compressed relay** cut inter-agent KV memory while **Judger-only concision steering** removes any downstream verbosity tax — *at iso-accuracy*?

Two orthogonal levers

H-OBF (memory). Compress each agent's relay KV: keep **4 sink** + **top-**k positions by importance (per layer); optional rank-8 low-rank "backfill" of the evicted mass.

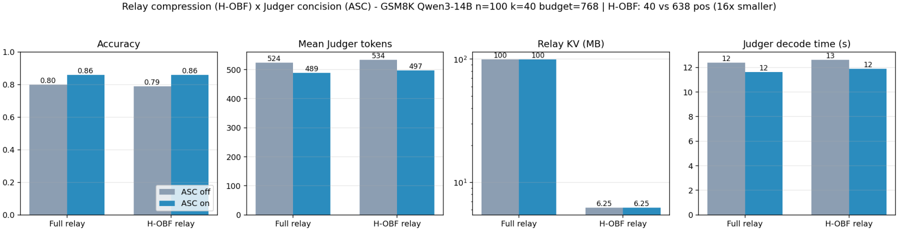
ASC (verbosity). A verbose→concise SEAL residual added at the **Judger decoding only** (layer 28, coef 40). Upstream agents left **unsteered**.

The 2×2 (paired, same items, greedy)

	No ASC	**ASC**
Full relay	A	B
H-OBF relay	C	D

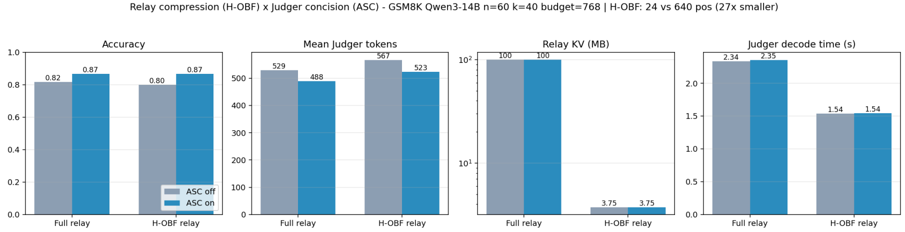
(+ plain-H eviction arms E/F as a backfill diagnostic)

Result 1 — moderate compression (budget 32, $n=100$)



- **Compression is free:** 16× less relay KV (100→6.25 MB, 638→40 pos), accuracy unchanged ($C-A=-0.01$, n.s.).
- **ASC and compression are orthogonal:** ASC cuts tokens the same on the compressed relay as the full one ($D-C=-37 \approx B-A=-36$); all interactions ≈ 0.
- **Best corner D** (compressed+ASC): +0.06 acc vs full, −27 tokens, 16× less KV.

Result 2 — aggressive compression (budget 16, n=60)



- **The verbosity tax now appears & is significant:** $C{-}A = +37$ tokens $[+11, +67]$ ($\sim +7\%$) — matching the compression paper's $\sim$10% claim.
- **ASC removes it:** $D{-}C = -44$ tokens $[-71, -16]$; D accuracy $= 0.867 =$ full+ASC.
- **Still near-free accuracy at 27×:** $C{-}A$ acc $= -0.017$ (n.s.).

Diagnostic — does the OBF backfill earn its keep?

arm	relay	acc	tokens	relay KV
A	full	0.817	529	100.0 MB (640 pos)
C	obf (rank-8 bf)	0.800	567	3.75 MB (24 pos)
E	**plain-H**	**0.817**	562	**2.50 MB (16 pos)**

- **Plain-H eviction beats OBF backfill:** E matches *full* accuracy (0.817) at **40**$\times$ **less KV**, using *less* memory than C.
- Our mean-pool rank-8 backfill adds off-manifold noise $\Rightarrow$ drop it (or replace with true low-rank factors).

Takeaways

Headline

Compress the relay, steer the reader. The inter-agent KV compresses **27–40$\times$ at iso-accuracy**; aggressive compression induces a small Judger verbosity tax that **Judger-only ASC fully absorbs**. Memory and verbosity are **decoupled**.

- **Memory** comes from compression; **tokens/latency/accuracy** come from ASC; they **stack**.
- **Plain headwise eviction** to 16 positions is the simple winner (40$\times$, no steering, full accuracy).
- Best combined corner: H-OBF + ASC = full+ASC accuracy at 27$\times$ less KV.

Method notes, limits, next

- **Faithfulness:** dense HF cache $\Rightarrow$ per-*layer* (not per-head-attention) selection; scorer is key-norm; pluggable.
- **Efficiency:** batched Judger decode $\Rightarrow$ GPU util $0 \to 60$–94%, $\sim 10\times$ faster (per-item timing CIs degenerate under batching — report throughput).
- **Limits:** single seed, GSM8K only, one relay-budget grid.

Next

budget-8 (find the accuracy cliff) · true low-rank backfill (beat plain-H?) · MedQA + second seed.